Section 4.5

0. Cauchy Integral Formula (Version #0) Let G be an open subset of $\mathbb{C}$ and let $f \in A(G)$. Let $a \in G$ and let R satisfy $B(a,R) \subset G$. Then , for z such that $|z-a| < r < R$ we have

$$f(z) = \frac{1}{2\pi i}\int_{|w-a|=r} \frac{f(w)}{w-z}\,dw$$

Cauchy's Theorem (Version #0) Let G be an open subset of $\mathbb{C}$ and let $f \in A(G)$. Let $a \in G$ and let R satisfy $B(a,R) \subset G$. Let γ be a closed rectifiable curve such $\{\gamma\} \subset B(a,R)$. Then,

$$\int_\gamma f = 0$$

1. Two approaches to extending the Cauchy Integral Formula and Cauchy's Theorem:

   a. Homology condition: $\gamma \approx 0$ wrt G, i.e., $n(\gamma,a) = 0$ for all $a \in \mathbb{C} \setminus G$

   b. Homotopy condition: $\gamma \sim 0$ (Section 4.6)

2. Let γ be a closed rectifiable curve in $\mathbb{C}$ and let $a \in \mathbb{C} \setminus \{\gamma\}$. Define

$$n(\gamma,a) = \frac{1}{2\pi i}\int_\gamma \frac{dw}{w-a}$$

   a. $n(\gamma,a)$ is continuous on components of $\mathbb{C} \setminus \{\gamma\}$

   b. $n(\gamma,a)$ is integer valued

   c. $n(\gamma,a)$ is 0 on the unbounded component of $\mathbb{C} \setminus \{\gamma\}$

3. Leibnitz's Rules

   a. Let $\varphi : [a,b] \times [c,d] \to \mathbb{C}$ such that φ is continuous on $[a,b] \times [c,d]$.

   Define $g(t) = \int_a^b \varphi(s,t)\,ds$. Then, g is continuous on [c ,d]. If

   $\varphi_2(s,t) = \frac{\delta}{\delta t}\varphi(s,t)$ is continuous on $[a,b] \times [c,d]$, then g' is

   continuously differentiable on [c ,d] and is given by $g'(t) = \int_a^b \varphi_2(s,t)\,ds$.

b. (Exercise 4.2.2) Let G be open and let γ be a closed rectifiable curve. Let $\varphi:\{\gamma\}\times G\to\mathbb{C}$ be continuous on $\{\gamma\}\times G$. Define $g(z)=\int\limits_{\gamma}\varphi(w,z)dw$. Then, g is continuous on G. If $\varphi_2(w,z)=\dfrac{\delta}{\delta z}\varphi(w,z)$ is continuous on $\{\gamma\}\times G$, then g is analytic on G and is given by $g'(z)=\int\limits_{\gamma}\varphi_2(w,z)dw$

4. Lemma (5.1) Let γ be a rectifiable curve and let φ be continuous on $\{\gamma\}$. Define $F_m(z)=\int_{\lambda}\dfrac{\varphi(w)}{(w-z)^m}dw$ for $z\in\mathbb{C}\setminus\{\gamma\}$. Then, $F_m\in A(\mathbb{C}\setminus\{\gamma\})$ and $F_m'=mF_{m+1}$.

5. Cauchy Integral Formula (Version #1) Let G be an open subset of $\mathbb{C}$ and let $f\in A(G)$. Let γ be a closed rectifiable curve in G such that $\gamma\approx 0$ wrt G. Then, for $z\in G\setminus\{\gamma\}$

$$n(\gamma,z)f(z)=\frac{1}{2\pi i}\int_{\gamma}\frac{f(w)}{w-z}dw$$

6. Cauchy's Theorem (Version #1) Let G be an open subset of $\mathbb{C}$ and let $f\in A(G)$. Let γ be a closed rectifiable curve in G such that $\gamma\approx 0$ wrt G. Then,

$$\int_{\gamma} f=0$$

7. Cauchy Integral Formula (Version #2) Let G be an open subset of $\mathbb{C}$ and let $f\in A(G)$. Let $\gamma_1,\gamma_2,\cdots,\gamma_m$ be closed rectifiable curves in G such that $\gamma_1+\gamma_2+\cdots+\gamma_m\approx 0$ wrt G. Then, for $z\in G\setminus\bigcup\limits_{k=1}^{m}\{\gamma_k\}$,

$$f(z)\sum_{k=1}^{m}n(\gamma_k,z)=\frac{1}{2\pi i}\sum_{k=1}^{m}\int_{\gamma_k}\frac{f(w)}{w-z}dw$$

8. Cauchy's Theorem(Version #2) Let G be an open subset of $\mathbb{C}$ and let $f\in A(G)$. Let $\gamma_1,\gamma_2,\cdots,\gamma_m$ be closed rectifiable curves in G such that $\gamma_1+\gamma_2+\cdots+\gamma_m\approx 0$ wrt G. Then,

$$\sum_{k=1}^{m} \int_{\gamma_k} f = 0$$

9. Corollary to CIF #1. Let G be an open subset of $\mathbb{C}$ and let $f \in A(G)$. Let γ be a closed rectifiable curve in G such that $\gamma \approx 0$ wrt G. Then, for $z \in G \setminus \{\gamma\}$ and $k > 0$

$$n(\gamma, z) f^{(k)}(z) = \frac{k!}{2\pi i} \int_{\gamma} \frac{f(w)}{(w-z)^{k+1}} dw$$

10. Corollary to CIF #2.) Let G be an open subset of $\mathbb{C}$ and let $f \in A(G)$. Let $\gamma_1, \gamma_2, \cdots, \gamma_m$ be closed rectifiable curves in G such that $\gamma_1 + \gamma_2 + \cdots + \gamma_m \approx 0$ wrt G. Then, for $z \in G \setminus \bigcup_{k=1}^{m} \{\gamma_k\}$ and $k > 0$

$$f^{(k)}(z) \sum_{k=1}^{m} n(\gamma_k, z) = \frac{k!}{2\pi i} \sum_{k=1}^{m} \int_{\gamma_k} \frac{f(w)}{(w-z)^{k+1}} dw$$

11. Morera's Theorem. Let G be a region. Let $f : G \to \mathbb{C}$ be continuous on G such that $\int_T f = 0$ for every triangular path T in G. Then, f is analytic on G.